



HANDS-ON LAB

# UI Modernisation Lab

## Maximo Real Estate & Facilities

*TRIRIGA Location Visualisation — Interactive Map Lab*

 Duration

**45–60 min**

 Level

**Beginner–Intermediate**

 Stack

**Python · HTML · TRIRIGA API**

### What You Will Build

-  Interactive US map visualising IBM TRIRIGA facility locations
-  Python data fetcher connecting to your TRIRIGA REST API
-  Filterable table view with CSV export capability
-  Customisable colour-coded markers by location type

# Lab Objectives & Architecture

*What you will learn and how the pieces fit together*

## Learning Objectives

1

### Connect to TRIRIGA REST API

Authenticate using JSESSIONID and call the locations endpoint to fetch facility data

2

### Process & Filter Location Data

Parse JSON responses, validate coordinates, and filter by country and data quality

3

### Build an Interactive Map

Use Leaflet.js to render colour-coded markers with clustering and popups

4

### Add Filtering & Export

Implement live dropdown filters and CSV export from the table view

5

### Customise the UI

Modify marker colours, map centre, and apply your own branding to the interface



## Architecture Overview

### IBM TRIRIGA Instance

*Your MREF environment*



### fetch\_tririga\_locations.py

*Python · REST API · JSON output*



### tririga\_us\_map.html

*Leaflet.js · Interactive Map*



### tririga\_location\_table.html

*Search · Filter · CSV export*

*Data Flow ↓*

# Step-by-Step Exercise Guide

Follow these steps in order — estimated 45 minutes total

## Phase 1

🕒 ~10 min

### Setup & Clone

1. Clone the repository from GitHub
2. Verify Python 3.10+ installed
3. Open `fetch_tririga_locations.py`
4. Set `TRIRIGA_BASE_URL` on line 21
5. Set username & password lines 22-23

```
git clone
https://github.com/
ibm-self-serve-assets/
UI-Modernisation-Lab-MREF
```

```
cd UI-Modernisation-Lab-
MREF
```

## Phase 2

🕒 ~10 min

### Fetch Data

1. Run the Python fetch script
2. Script authenticates via JSESSIONID
3. Fetches all location records from TRIRIGA
4. Filters US locations with valid GPS
5. Creates 3 output files (JSON + CSV)

```
python3 fetch_tririga_
locations.py

# Output:
# tririga_locations.json
```

```
# tririga_us_locations.json
```

## Phase 3

🕒 ~10 min

### Launch Map

1. Start the local web server
2. Open the map URL in browser
3. Verify markers appear on US map
4. Click markers to see location details
5. Try the collapsible control panel

```
python3 -m http.server
8000

# Then open:
# http://localhost:8000/
```

```
# tririga_us_map.html
```

## Phase 4

🕒 ~15 min

### Customise & Explore

1. Change marker colours by type
2. Adjust map default centre/zoom
3. Add a new filter dropdown
4. Open table view & test CSV export
5. Try filtering different country

```
// Edit
tririga_us_map.html
const getMarkerColor =
(locationType) => {
  if (type.includes
    ('building'))
    return '#c30b82';
```



Tip: If no locations appear on the map — check that `tririga_us_locations.json` exists and has records. Press F12 to open browser console for error details.

# Key Features, Troubleshooting & Next Steps

*What the lab produces — and where to take it further*



## Interactive Map

Leaflet.js map with OpenStreetMap tiles, smart marker clustering, and popup details



## Advanced Filtering

Live dropdowns filter by state, city and location type with real-time count updates



## Live Statistics

Real-time breakdown panel showing counts by type and total filtered locations



## CSV Export

One-click export of filtered results to CSV for use in Excel or Power BI



## Common Issues & Fixes

- No locations on map:** Run Python script first; check JSON file exists
- Authentication fails:** Verify URL has no trailing /app/tririga; check credentials
- All coordinates zero:** Enable GIS fields in TRIRIGA admin; re-fetch
- CORS error in browser:** Use `python3 -m http.server` — do not open HTML directly
- Timeout errors:** Script retries 3× automatically; check VPN/standby



## Next Steps & Extensions

1

### Add Heatmap Layer

Visualise density of incidents or assets

2

### Multi-Country Support

Extend filter to global TRIRIGA locations

3

### Real-time Data Refresh

Auto-fetch every N minutes without reload

4

### Dark Mode Theme

Toggle between light and dark map styles

5

### Mobile App Version



Star & fork the repo: [github.com/ibm-self-serve-assets/UI-Modernisation-Lab-MREF](https://github.com/ibm-self-serve-assets/UI-Modernisation-Lab-MREF) · Questions? [ssnarasi@in.ibm.com](mailto:ssnarasi@in.ibm.com)